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JAVA PROJECT REPORT ON

PERSONAL EXPENSE TRACKING SYSTEM

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Academic Year: **2025 – 2026**
APRIL 2026
Semester: **I**

DECLARATION

I hereby declare that the project entitled "PERSONAL EXPENSE TRACKING SYSTEM" submitted in partial fulfillment of the requirements for the course [DHANUSHYA/B.SC.IT] is a

record of my original work carried out under the guidance of [DHILEEBAN A], Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore DHANUSHYA M

CERTIFICATE

This is to certify that the project entitled "PERSONAL EXPENSE TRACKING SYSTEM" is a bona fide work carried out by [DHANUSHYA] [B.SC.IT-A /], [Department Name], KG College of Arts & Science, in partial fulfillment of the requirements for the course [JAVA PROGRAMMING].

This project has been completed under my supervision and guidance during the academic period April 2026.

Place:

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:Head of the Department
(Technical Trainer, KGCAS)(Technical Head, KGCAS)

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ABSTRACT

The Personal Expense Tracking System is a Java-based application designed to help individuals manage their daily income and expenditure effectively. In many households and among students and working professionals, tracking personal finances is often done manually using notebooks or spreadsheets, which can be error-prone and difficult to maintain. This project aims to simplify personal financial management by providing an automated system that records income and expense transactions, categorizes them, and displays a summary of the financial status.

The system allows users to add expense entries under various categories such as Food, Transport, Education, Entertainment, and Utilities. It also allows income entries to be recorded and calculates the remaining balance automatically. The application is developed using Core Java programming concepts, including classes, objects, methods, arrays, conditional statements, loops, and exception handling. The project can be implemented as a console-based application or enhanced with a Java Swing graphical user interface.

The main objective of the system is to help users monitor their spending habits, control unnecessary expenses, and maintain a healthy financial balance. This project demonstrates the practical implementation of object-oriented programming concepts in Java and provides a strong foundation for developing advanced personal finance management applications.

1. INTRODUCTION

In today's fast-paced world, managing personal finances effectively is a critical life skill. With rising living costs and multiple sources of expenses, it becomes increasingly difficult for individuals to keep track of where their money is being spent. Most people lack a systematic approach to monitoring their daily income and expenditure, which often results in overspending and financial instability.

Traditionally, people maintained hand-written records of their expenses in notebooks or diaries. While simple, this method is time-consuming, prone to calculation errors, and difficult to organize or retrieve. The need for a more efficient, automated solution has become evident.

The Personal Expense Tracking System is a Java-based application that provides an organized and automated method for recording, categorizing, and summarizing financial transactions. The system enables users to add income and expense entries, view a category-wise breakdown of spending, and check the available balance at any time.

This project is developed using Core Java programming concepts such as classes, objects, arrays, loops, conditional statements, and methods. It can be implemented as a console-based program or extended with a Java Swing GUI for a more interactive experience.

The primary goal of this system is to promote financial discipline, reduce human error in manual bookkeeping, and give users clear visibility into their spending patterns. This project also demonstrates how Java programming can be applied to develop real-world personal productivity applications.

1.1 Objective

The main objective of the Personal Expense Tracking System is to develop a simple and efficient Java-based application that automates the recording and monitoring of personal income and expenses.

The specific objectives of this project are:

- To design and develop a Java-based personal expense tracking application.
- To allow users to record daily income and expense transactions.
- To categorize expenses under predefined categories such as Food, Transport, Education, and Entertainment.
- To automatically calculate total income, total expenses, and available balance.
- To provide a category-wise summary of expenses for better financial awareness.
- To demonstrate the implementation of Core Java programming concepts such as classes, objects, arrays, loops, and methods.
- To generate a clear and organized financial report for the user.

1.2 Overview

The Personal Expense Tracking System is a menu-driven application developed using the Java programming language. The system provides a structured platform where users can record their daily financial transactions, view spending summaries, and monitor their remaining balance.

The system displays a menu with options to add income, add an expense under a selected category, view all transactions, view a category-wise expense report, and check the current balance. The user can interact with the system by selecting options from the menu and entering the required details.

The application automatically calculates the total income, total expenses, and current balance based on the transactions entered. It also groups expenses by category and displays a summary to help users understand where their money is being spent.

This project demonstrates the practical use of Java programming in developing real-world productivity tools for personal financial management.

1.3 Features

The Personal Expense Tracking System includes the following main features:

- Add Income – Allows the user to record income entries with a description and amount.
- Add Expense – Allows the user to record expense entries under selected categories such as Food, Transport, Education, Entertainment, and Utilities.
- View All Transactions – Displays a complete list of all recorded income and expense entries.
- Category-wise Expense Summary – Groups and displays total spending under each expense category.
- Balance Calculation – Automatically calculates and displays the available balance (Total Income − Total Expenses).
- User-Friendly Menu – Provides a simple and easy-to-navigate menu-driven interface.
- Input Validation – Handles invalid or incorrect inputs gracefully using exception handling.
- Fast and Accurate – Eliminates manual calculation errors and provides instant results.

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in manual expense tracking and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

Many individuals, especially students and young professionals, find it difficult to track their personal expenses on a daily basis. The reliance on manual methods such as writing in notebooks or using spreadsheets without automation often leads to several problems.

Some common problems include:

- Errors in manual addition and subtraction of income and expenses.
- Difficulty in categorizing and reviewing spending patterns.
- Lack of real-time visibility into available balance.
- Time-consuming process of manually organizing and searching records.
- No automated alerts or summaries for overspending in a particular category.

These challenges make it hard for individuals to maintain financial discipline and make informed spending decisions.

Therefore, there is a need for an automated expense tracking system that can record transactions quickly, categorize expenses, calculate the balance automatically, and present a clear summary of the financial status.

The Personal Expense Tracking System is designed to solve these problems by providing a computerized solution that automates all aspects of personal financial record-keeping.

2.2 Project Overview

The Personal Expense Tracking System is a Java-based application developed to help individuals manage their daily income and expenses in an organized and efficient manner.

The system provides a menu-driven interface where users can perform the following operations:

- Add a new income entry with a description and amount.
- Add a new expense entry by selecting a category and entering the amount.
- View all recorded transactions in a structured list.

- View a category-wise summary showing total spending per category.
- Check the current available balance.

This project is implemented using Core Java programming concepts and can be further developed into a Java Swing GUI application for a more user-friendly experience.

2.3 Scopes and Limitations

Scope

The scope of the Personal Expense Tracking System includes:

- Recording personal income and expense transactions.
- Categorizing expenses into predefined categories.
- Displaying a summary of all financial transactions.
- Calculating total income, total expenses, and available balance.
- Demonstrating the practical application of Core Java programming concepts.

The system can be used by:

- Students managing monthly allowances and educational expenses.
- Working professionals tracking salary and living expenses.
- Households monitoring daily household expenditures.
- Freelancers tracking project income and business-related expenses.

Limitations

- The system does not store data permanently (no file or database storage in the basic version).
- It does not provide graphical charts or visual reports in the console version.
- The system does not support multiple user accounts.
- Budget limit alerts and notifications are not included in the basic version.

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The Personal Expense Tracking System is designed as a menu-driven application where users can add income and expense entries, view transactions, and check the balance at any time.

The system is divided into several modules, each responsible for a specific function.

3.1 Modules

The main modules of the Personal Expense Tracking System are explained below:

1. Main Menu Module

This module is the entry point of the application. It displays the main menu with all available options and accepts the user's choice to navigate to the required function.

Menu Options:

- 1. Add Income
- 2. Add Expense
- 3. View All Transactions
- 4. View Category-wise Summary
- 5. Check Balance
- 6. Exit

2. Income Module

This module handles the recording of income transactions. The user enters a description (e.g., "Monthly Salary", "Freelance Payment") and the amount received. The entry is saved and the total income is updated.

3. Expense Module

This module allows the user to record expense transactions. The user selects an expense category from the following list:

- Food & Dining
- Transport
- Education

- Entertainment
- Utilities & Bills
- Others

The user then enters a description and the amount spent. The total expenses and category totals are updated accordingly.

4. Transaction History Module

This module displays a complete list of all recorded transactions, showing the type (Income/Expense), category, description, and amount for each entry.

5. Category Summary Module

This module groups all expenses by category and displays the total amount spent in each category. This helps users identify which categories consume the most of their budget.

6. Balance Module

This module calculates and displays the current balance using the formula:

```
Available Balance = Total Income – Total Expenses
```

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Personal Expense Tracking System, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1 Problem Analysis

In this phase, the problem of manual personal expense tracking is studied and analyzed in detail. The requirements of the system are identified based on the needs of individuals who wish to manage their daily finances.

The main problems identified in manual expense tracking are:

- Manual calculations may lead to errors in determining total income, expenses, and balance.
- Organizing and retrieving expense records manually is time-consuming.
- Difficulty in understanding spending patterns without a category-wise summary.
- Lack of a real-time view of available balance makes budgeting difficult.

Based on these problems, the system requirements were defined to develop an automated expense tracking application.

4.2 System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

- Main Menu Module
- Income Module
- Expense Module
- Transaction History Module
- Category Summary Module

- Balance Module

The program flow and structure are designed so that users can easily interact with the system through a simple menu.

4.3 Program Development

After designing the system, the program is implemented using the Java programming language.

The following steps were followed during development:

- Creation of a Transaction class to represent income and expense entries.
- Implementation of the menu-driven main program loop.
- Writing methods for adding income, adding expenses, viewing transactions, and displaying balance.
- Handling user input using Scanner class.
- Implementing exception handling for invalid inputs.

Core Java concepts such as classes, objects, arrays, ArrayList, loops, conditional statements, and methods were used during development.

4.4 Testing and Validation

Testing ensures that the program works correctly and produces the expected results for various inputs.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	Expected Result
Add income of ₹5000	Income recorded, balance updated to ₹5000
Add food expense of ₹500	Expense recorded, balance updated to ₹4500
View all transactions	All income and expense entries displayed
View category summary	Total per category shown correctly
Enter invalid amount (text)	Error message displayed, program continues
Check balance after entries	Correct balance (Income − Expenses) shown

This phase ensures that the program runs without errors and produces accurate results.

4.5 Result and Documentation

After successful testing, the results of the system are documented. The system correctly records income and expense transactions, calculates the balance, and displays a category-wise summary.

The results are demonstrated through program execution and screenshots showing the input and output of the system.

Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6 Algorithm

Algorithm for the Personal Expense Tracking System:

1. Start the program.
2. Display the main menu with options: Add Income, Add Expense, View Transactions, View Summary, Check Balance, Exit.
3. Accept the user's menu choice.
4. If the user selects Add Income: Accept description and amount, save the income entry, update total income.
5. If the user selects Add Expense: Display expense categories, accept category selection, description and amount, save expense entry, update total expenses and category total.
6. If the user selects View Transactions: Display all recorded transactions with type, category, description, and amount.
7. If the user selects View Category Summary: Group expenses by category and display total spending per category.
8. If the user selects Check Balance: Calculate Balance = Total Income − Total Expenses, display the result.
9. If the user selects Exit: Terminate the program.
10. Repeat from Step 2 until the user chooses Exit.
11. End the program.

4.7 Flowchart

Students should draw a flowchart showing the program flow for the Personal Expense Tracking System.

Example flow:

- Start
- Display Main Menu
- Accept User Choice
- Choice = 1 → Add Income → Update Balance
- Choice = 2 → Select Category → Add Expense → Update Balance
- Choice = 3 → Display All Transactions

- Choice = 4 → Display Category Summary
- Choice = 5 → Display Current Balance
- Choice = 6 → Exit
- Loop back to Menu until Exit is selected

Students can draw the flowchart using:

- draw.io
- Lucidchart
- MS Visio

5. SOURCE CODE

```
package expense;

import java.util.Scanner;

public class MainApp {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        ExpenseService service = new ExpenseService();
        User currentUser = null;

        int choice;

        do {
            System.out.println("\n1. Register");
            System.out.println("2. Login");
            System.out.println("3. Exit");

            choice = sc.nextInt();
            sc.nextLine();

            switch (choice) {

                case 1:
                    System.out.print("Enter name: ");
                    String name = sc.nextLine();
                    System.out.print("Enter monthly budget: ");
                    double budget = sc.nextDouble();
                    service.registerUser(name, budget);
                    break;

                case 2:
                    System.out.print("Enter name: ");
                    String loginName = sc.nextLine();
                    currentUser = service.login(loginName);

                    if (currentUser != null) {
                        int ch;
                        do {
                            System.out.println("\n1. Add Expense");
                            System.out.println("2. View Expenses");
                            System.out.println("3. Summary");
                            System.out.println("4. Logout");

                            ch = sc.nextInt();
```

```
                    switch (ch) {
                        case 1:
                            sc.nextLine();
                            System.out.print("Category: ");
                            String cat = sc.nextLine();
                            System.out.print("Amount: ");
                            double amt = sc.nextDouble();
                            service.addExpense(currentUser.getName(), cat, amt);
                            break;

                        case 2:
                            service.showExpenses(currentUser.getName());
                            break;

                        case 3:
                            service.showSummary(currentUser);
                            break;
                    }

                } while (ch != 4);
            }
            break;
        }

    } while (choice != 3);

    sc.close();
  }
}


package expense;

import java.util.*;

public class ExpenseService {

  private HashMap<String, User> users = new HashMap<>();
  private HashMap<String, ArrayList<Expense>> expenses = new HashMap<>();

  public void registerUser(String name, double budget) {
    users.put(name, new User(name, budget));
    expenses.put(name, new ArrayList<>());
    System.out.println("User Registered!");
  }

  public User login(String name) {
    if (users.containsKey(name)) {
      return users.get(name);
```

```
        }
        System.out.println("User not found!");
        return null;
    }

    public void addExpense(String name, String category, double amount) {
        Expense exp = new Expense(category, amount);
        expenses.get(name).add(exp);
        users.get(name).addExpense(amount);
        System.out.println("Expense Added!");
    }

    public void showExpenses(String name) {
        for (Expense e : expenses.get(name)) {
            System.out.println(e.getCategory() + " - " + e.getAmount());
        }
    }

    public void showSummary(User user) {
        System.out.println("Total Expense: " + user.getTotalExpense());
        System.out.println("Remaining Budget: " + user.getRemainingBudget());
    }
}
package expense;

public class User {
    private String name;
    private double monthlyBudget;
    private double totalExpense;

    public User(String name, double monthlyBudget) {
        this.name = name;
        this.monthlyBudget = monthlyBudget;
        this.totalExpense = 0;
    }

    public String getName() {
        return name;
    }

    public double getTotalExpense() {
        return totalExpense;
    }

    public void addExpense(double amount) {
        totalExpense += amount;
    }

    public double getRemainingBudget() {
```

```
        return monthlyBudget - totalExpense;
    }
}

package expense;

public class Expense {
    private String category;
    private double amount;

    public Expense(String category, double amount) {
        this.category = category;
        this.amount = amount;
    }

    public String getCategory() {
        return category;
    }

    public double getAmount() {
        return amount;
    }
}
```

6. INPUT AND OUTPUT SCREENS

```
=== PERSONAL EXPENSE TRACKER ===
1. Add Income
2. Add Expense
3. View All Transactions
4. View Category Summary
5. Check Balance
6. Exit
Enter your choice: _
```

7. DISCUSSION & CONCLUSION

7.1 Discussion

The Personal Expense Tracking System was developed to automate the process of recording and monitoring personal income and expenses. The system allows users to add income and expense transactions, categorize spending, and view a clear financial summary at any time.

During the development process, Core Java programming concepts such as classes, objects, ArrayList, loops, conditional statements, methods, and exception handling were applied effectively. The system successfully performs income and expense tracking, calculates the balance automatically, and presents a structured category-wise report.

The system was tested with various input scenarios to verify that all modules function correctly. Test results confirm that the system records transactions accurately, handles invalid inputs without crashing, and computes the correct balance and category totals.

The project demonstrates that Java programming can be used to develop practical, real-world personal finance tools that simplify everyday tasks and promote better financial management.

7.2 Conclusion

The Personal Expense Tracking System successfully demonstrates the practical application of Core Java programming concepts in developing a useful personal finance management application.

The system automates the recording of income and expenses, reduces the chance of calculation errors, and provides users with clear visibility into their spending habits. By categorizing expenses and displaying a running balance, the system empowers users to make more informed financial decisions and maintain better control over their budget.

This project helps students understand how Java programming can be applied to develop practical software tools. The project can be further enhanced by adding features such as file-based or database storage for persistent records, a graphical user interface using Java Swing, monthly budget setting and alerts, export to CSV for reporting, and multi-user account support.

8. REFERENCES

The following resources were used during the development of this project:

Books

- Herbert Schildt – Java: The Complete Reference
- Cay S. Horstmann – Core Java Volume I – Fundamentals
- E. Balagurusamy – Programming with Java

Websites

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- https://www.geeksforgeeks.org
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- https://www.javatpoint.com
- https://docs.oracle.com/javase/tutorial